

● EASY

● ADVANCED MATH

Advanced Math

06

10 questions · ~15 min

Q1

ADVANCED MATH

$$x + 7 = 10$$

$$x + 7^2 = y$$

Which ordered pair x, y is a solution to the given system of equations?

A. 3, 100

B. 3, 3

C. 3, 10

D. 3, 70

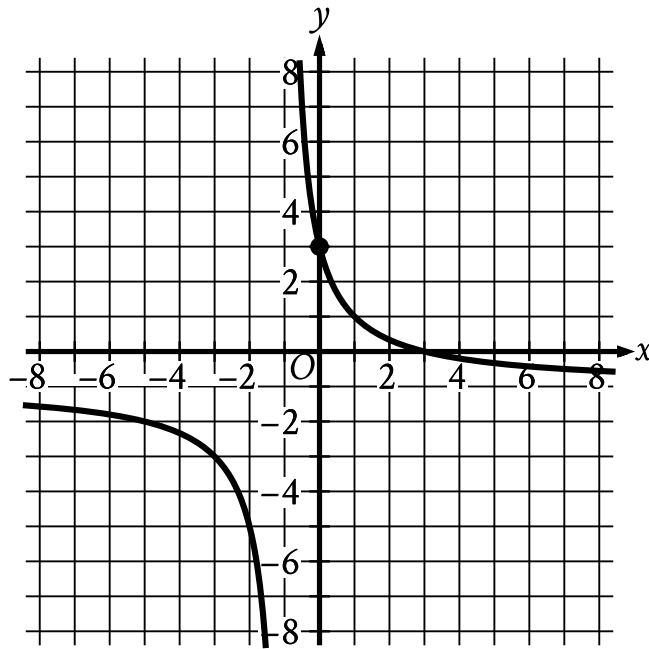
Which expression is equivalent to $256w^2 - 676$?

A. $(16w - 26)(16w - 26)$

B. $8w - 138w + 13$

C. $(8w - 13)(8w - 13)$

D. $16w - 2616w + 26$



- For the first curve:
 - Moving from left to right, the curve is in quadrant 3.
 - As x decreases, the curve approaches the line y equals negative 1.
 - As x increases, the curve approaches the line x equals negative 1.
- For the second curve:
 - Moving from left to right, the curve passes from quadrant 2 to quadrant 1 to quadrant 4.
 - As x decreases, the curve approaches the line x equals negative 1.
 - As x increases, the curve approaches the line y equals negative 1.
- The 2 curves pass through the following points:
 - (negative 6 comma negative nine fifths)
 - (0 comma 3)

- What is the y -coordinate of the y -intercept of the graph shown?

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

ADVANCED MATH — SET 6 · ADVANCED MATH

Q4

GRID-IN

ADVANCED MATH

$$5x = 45$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression is equivalent to $12x^3 - 5x^3$?

A. $7x^6$

B. $17x^3$

C. $7x^3$

D. $17x^6$

An investment account was opened with an initial value of \$ 890. The value of the account doubled every 10 years. Which equation represents the value of the account Mt , in dollars, t years after the account was opened?

A. $Mt = 890 \frac{1}{2}^{\frac{t}{10}}$

B. $Mt = 890 \frac{1}{10}^{\frac{t}{2}}$

C. $Mt = 890 2^{\frac{t}{10}}$

D. $Mt = 890 10^{\frac{t}{2}}$

Q7

GRID-IN) ADVANCED MATH

$$\sqrt{x+4} = 11$$

What value of x satisfies the equation above?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression is equivalent to $9x^3 + 5x + 7 + 6x^3 + 5x^2 - 5$?

- A. $15x^6 + 5x^2 - 5x - 35$
- B. $15x^3 + 10x^2 + 2$
- C. $15x^6 + 5x^2 + 5x + 2$
- D. $15x^3 + 5x^2 + 5x + 2$

The function g is defined by $g(x) = |x + 18|$. What is the value of $g(4)$?

A. -18

B. -4

C. 14

D. 22

$$\frac{r}{38} = s + t$$

The given equation relates the quantities r , s , and t . Which equation correctly expresses r in terms of s and t ?

A. $r = s + t + 38$

B. $r = s + t - 38$

C. $r = 38s + t$

D. $r = \frac{s+t}{38}$